

Title: An Intelligent Framework for Detecting, Analysing, and Revocation of Malicious URLs and HTML-Based Web Attacks.

Domain: Machine Learning

Abstract :

Malicious URLs and compromised web pages are one of the major sources of cyberattacks such as phishing, malware delivery, and data theft. Attackers frequently design URLs that appear legitimate and host malicious content inside web pages, making detection difficult using traditional security methods. Existing systems mainly depend on blacklists and predefined signatures, which are ineffective against newly emerging and zero-day attacks.

This project proposes an **AI-based web security system** that follows a **three-stage detection process**. First, the system analyses and classifies the URL to identify whether it is suspicious or malicious. Second, if the URL is accessible, the corresponding web page is fetched and its HTML content is scanned for hidden malicious scripts and abnormal behaviours. Finally, if the URL or web page is confirmed to be malicious, the system automatically **blocks the URL and revokes access** by adding it to a blacklist. The proposed system improves detection accuracy, reduces false positives, and enhances web security.

Technical Justification of the Title

1. Problem background and motivation

With the rapid growth of internet usage, users frequently access web links through emails, messages, and social media. Many of these links lead to malicious websites designed to steal sensitive information or infect systems. Traditional security mechanisms fail to analyze both the URL structure and the internal content of web pages effectively. This motivates the need for an intelligent system that detects threats at multiple levels before allowing access.

2. Existing System

The existing systems use:

- Blacklist-based URL filtering
- Signature-based malware detection □ Static rule-based security checks

These systems rely on previously known attack patterns.

Drawbacks of Existing System

- Cannot detect **zero-day malicious URLs**
- High **false positive and false negative rates**
- No deep analysis of web page HTML content
- Manual updates required frequently

3. Need and relevance of the proposed project

The proposed project is needed to overcome the limitations of existing systems by combining URL-level detection with web page-level analysis. It is highly relevant for protecting users from modern cyber threats where attackers hide malicious content inside seemingly safe URLs.

Project Objectives

The proposed project aims to:

1. Detect and classify URLs using machine learning techniques.
2. Analyse web page HTML content for malicious scripts and behaviours.
3. Reduce false positives using multi-level detection logic.
4. Detect unknown and zero-day malicious URLs.
5. Block and revoke access to confirmed malicious URLs automatically.

Project Scope & Limitations

Scope:

- Detection of malicious URLs and malicious web pages
- Application-layer web security
- Suitable for academic, enterprise, and IoT-based environments

Limitations (if any):

- Detection accuracy depends on training dataset quality
- Limited real-time behavioural execution analysis
- Large-scale deployment requires additional optimization

Proposed Technical Approach

Brief description of the approach to be adopted:

1. Overall System / Model Description

The system accepts a URL from the user and performs an initial URL analysis using a trained machine learning model. If the URL is reachable, the system retrieves the corresponding web page and analyzes its HTML content for malicious patterns. A decision engine combines the results from both stages to classify the URL as safe or malicious. If malicious, the URL is blocked and added to a blacklist.

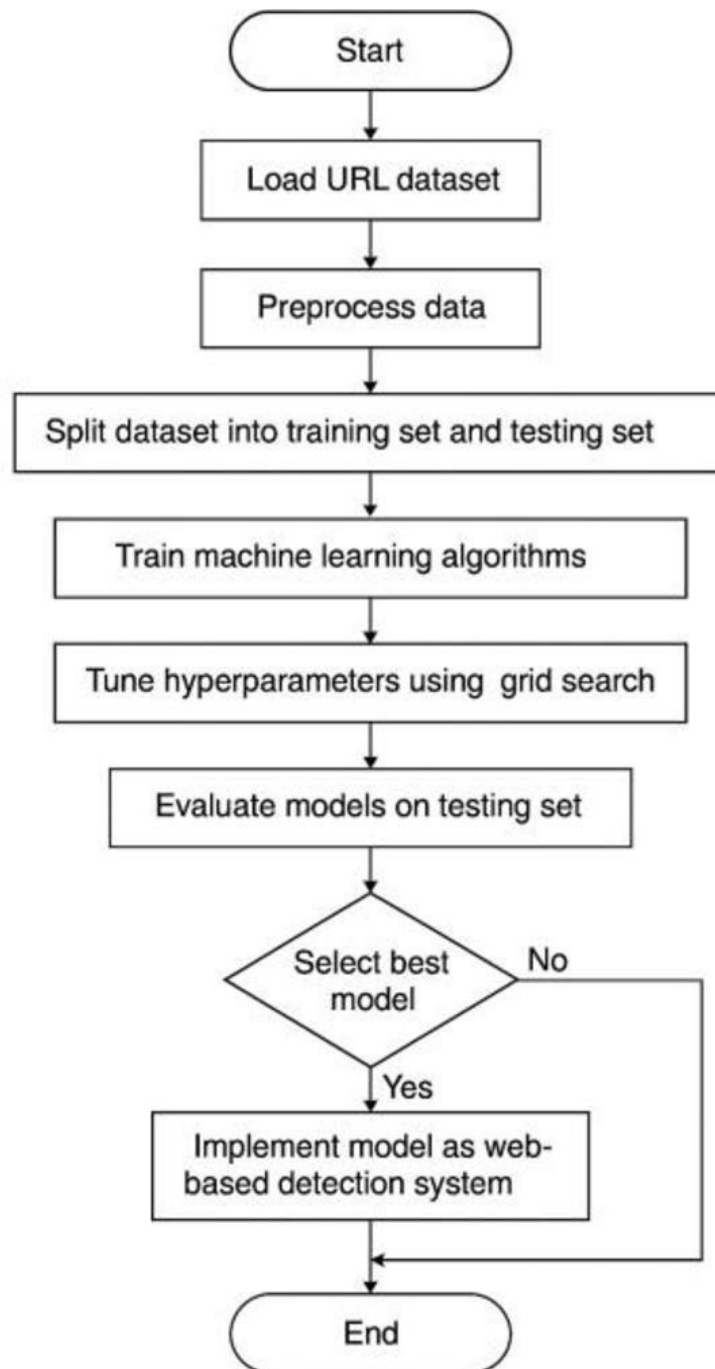
2. Algorithms / Frameworks / Methodologies

- Supervised machine learning algorithms for URL classification
- URL feature extraction techniques
- HTML content parsing and pattern analysis
- Rule-based and decision-based classification logic

3. Tools, Software, and Platforms

- Frontend: HTML, CSS
- Backend: Python (Flask framework)
- Machine Learning Libraries: Scikit-learn, Pandas, NumPy
- Development Tools: Visual Studio Code
- Operating System: Windows / Linux

FLOWCHART



Resource & Feasibility Assessment

Resource	Availability
Required Software	Available
Required Hardware	Available
Data / Dataset	☐ Available ☐ To be Created
Overall Feasibility	☐ Feasible ☐ Needs Revision

Declaration by Student(s)

We certify that the proposed title is original and the project will be carried out in compliance with Anna University rules and ethical standards.

Signature of Student(s):

1.

2.

Date: _____

Recommendation of Project Guide / Supervisor

☐ Recommended ☐ Recommended with Modification ☐ Not Recommended

Name and Designation of Guide :

Signature : _____

Date : _____

Approval of Project Review Committee

Role	Name	Signature	Date
Project Supervisor			
Project Coordinator			

Remarks (if any)

Approval of Head of the Department

☐ Approved ☐ Not Approved

Name & Signature of HOD: Dr.R.VENKATARATHINAM -

Date: